

- 1983/1. Find all functions f defined on the set of positive real numbers which take positive real values and satisfy the conditions:

$$f(xf(y)) = yf(x)$$

for all positive real x, y ;

$$f(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow \infty$$

- 1983/2. Let A be one of the two distinct points of intersection of two unequal coplanar circles C_1 and C_2 with centers O_1 and O_2 respectively.

One of the common tangents to the circles touches C_1 at P_1 and C_2 at P_2 , while the other touches C_1 at Q_1 and C_2 at Q_2 .

Let M_1 be the midpoint of P_1Q_1 and M_2 be the midpoint of P_2Q_2 .

Prove that

$$\angle O_1AO_2 = \angle M_1AM_2$$

- 1983/3. Let a, b, c be pairwise relatively prime positive integers.

Prove that

$$2abc - ab - bc - ca$$

is the largest integer which cannot be expressed in the form

$$xbc + yca + zab$$

where x, y, z are non-negative integers.

1983/4. Let E be a set of points on the sides of an equilateral triangle ABC , such that its distance from each of the vertices A, B, C is a rational number.

Prove that there exist points P, Q, R in E such that PQR is an equilateral triangle.

1983/5. Is it possible to choose 1983 distinct positive integers, all not exceeding 10^5 , no three of which form an arithmetic progression?

1983/6. Let a, b, c be the lengths of the sides of a triangle.

Prove that

$$a^2b(a-b) + b^2c(b-c) + c^2a(c-a) \geq 0$$